# ERD/ERS, MI & ErrP — Training Methodology

Two-model EEG paradigm: 3-class motor imagery (rest / left / right) with the ERD/ERS gate built in, plus error-related potentials

biopipe / Hybrid EEG–EMG BMI • OpenBCI Cyton (8 ch, 250 Hz) + Ultracortex • Montage Fz, FC5, FC6, C3, Cz, C4, P3, P4 (A1 ref, A2 bias) • Revision 2, 2 October 2026 — replaces the 21 September 2026 three-model version (changes listed at the end)

## 1. Overview

The EEG side trains exactly **two models**:

- **MI model — rest / mi_left / mi_right.** One 3-class classifier. Its *rest* class is the ERD/ERS gate: sensorimotor mu/beta rhythms are strong at rest and desynchronise during imagery, so “rest vs imagery” is the gate and “left vs right” is the command — learned together, with no separate gate model.
- **ErrP model — correct / error.** Detects the brain’s response to a wrong outcome so the controller can veto or undo an action.

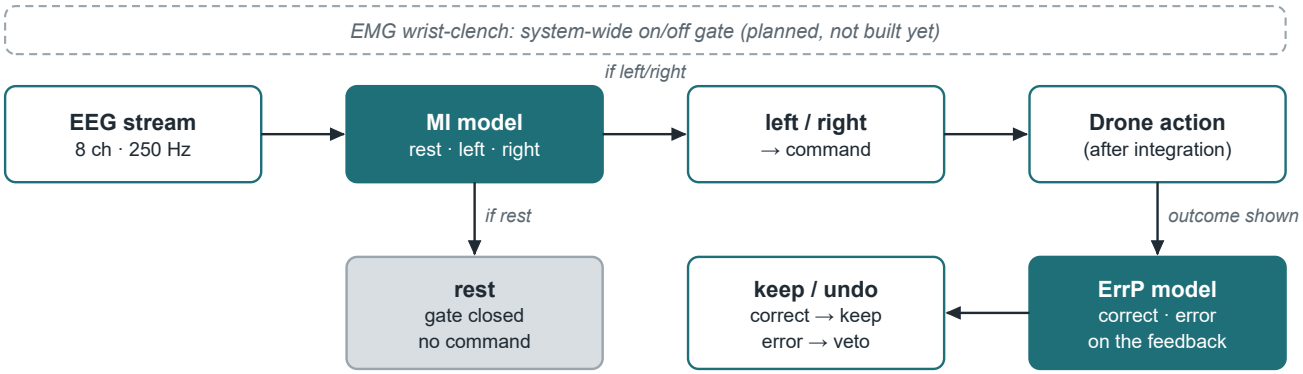



*Figure 1 — Control loop. A rest prediction closes the gate; a left/right prediction becomes a command, and the ErrP model checks the brain’s response to its outcome.*

## 2. Montage

Physically available holes only (FCz, FC3/FC4, CP3/CP4 and CPz are unavailable). Load it with **Config → Load MI/ErrP preset**. Standard OpenBCI EEG wiring: electrodes in the bottom-row inputs N1P–N8P, reference ear clip in SRB2, bias ear clip in BIAS.

**Table 1 — Electrode map**

| Electrode | GUI ch | Cyton input | Used for |
|---|---|---|---|
| **Fz** | 0 | N1P (ch 1) | ErrP fronto-central (ERN/FRN); blink check |
| **FC5** | 1 | N2P (ch 2) | MI — left premotor/hand area (CSP) |
| **FC6** | 2 | N3P (ch 3) | MI — right premotor/hand area (CSP) |
| **C3** | 3 | N4P (ch 4) | MI — left hand area: ERD during *right*-hand imagery |
| **Cz** | 4 | N5P (ch 5) | MI midline; ErrP central (ERN, Pe) |
| **C4** | 5 | N6P (ch 6) | MI — right hand area: ERD during *left*-hand imagery |
| **P3** | 6 | N7P (ch 7) | MI posterior; ErrP Pe; eyes-closed alpha check |
| **P4** | 7 | N8P (ch 8) | MI posterior; ErrP Pe; eyes-closed alpha check |
| **A1 (left ear)** | ref | SRB2 | Reference — required |
| **A2 (right ear)** | bias | BIAS | Bias (driven ground) — required |

## 3. Shared EEG settings (Config and Monitor tabs)

These apply to both models. Notch, band limits and common-average reference change only what the Monitor shows: recordings are saved **raw**, and each model applies its own band-pass when it trains and runs (Appendix A).

**Table 2 — Config and Monitor settings**

| Tab | Setting | Recommended | Options / range | Affects |
|---|---|---|---|---|
| **Config** | Profile | One per setup, e.g. *MI-8ch* | Load · Delete · Save as | Restores every Config setting below in one click. |
| | Source | cyton (synthetic for a dry run) | cyton · lsl · synthetic | Where samples come from. |
| | Cyton port | The dongle's port | e.g. COM3 | Device Manager → Ports. Close the OpenBCI GUI first. |
| | Sample rate | 250 | Hz | Must match the board (Cyton: 250 Hz). |
| | Montage (10-20 head) | Load MI/ErrP preset | Click sets Data · Ref (SRB) · Bias · Clear; channel 0–7 per Data node | Electrode→channel map saved with every recording. Live and pooling need the same map. |
| | Actions | mi_left, mi_right | Add · Remove | The two MI classes (rest is added automatically). Names containing "left"/"right" set the cue arrows. Keep exactly these two. |
| | Notch Hz | 50 | Hz | Monitor only (India mains). |
| | Band low / high | 1 / 40 (high 45–50 to see γ) | Hz | Monitor only. |
| | Common-avg ref | On | On · off | Monitor only. |
| | Window ms | 1000–2000 | ms | Monitor band-power window. |
| | Step ms | 250 | ms (Live minimum 150) | Monitor refresh and Live update rate. |
| | Alpha thresh | ≈ 0.2, then calibrate | 0–1 (alpha share) | Alpha light; electrodes set to the most posterior pair (P3/P4) on Apply. |
| | Apply & Save | After any change | — | Writes config/eeg.yaml and stops the stream — press Start again. |
| **Monitor** | alpha test | On | Checkbox | Lights when P3/P4 alpha share exceeds Alpha thresh. |
| | waterfall | On during checks | Checkbox | Spectrogram: frequency across (0 → Band high), time downward. |
| | simulate eyes-closed | — | Synthetic source only | Injects alpha to test the pipeline without hardware. |

## 4. Headset setup and signal quality (every session)

- **Connect reference and bias.** A1 → SRB2, A2 → BIAS (never AVDD/AVSS/AGND). Without both you record mains hum and drift, not EEG — the giveaway is a band composition swamped by δ.
- **Contact.** Part the hair and twist each comb down to the scalp. Aim for impedances in the low tens of kΩ and similar across channels; mismatched or very high contacts let mains leak in, and common-average reference spreads one bad channel into all of them.
- **Monitor checks** (EEG → Monitor → Start): no channel flagged **RAILED**, **FLAT** or *near rail* — judge by the flags, not the traces: with common-average reference on, a dead channel still draws a convincing trace. A blink gives a slow deflection at Fz; a jaw clench gives a broadband burst on all channels that settles back; the band composition shows a spread of θ/α/β rather than a wall of δ.
- **Eyes-closed test.** 15 s eyes open, 15 s closed: alpha should rise at P3/P4 (alpha light, or a bright ~10 Hz column on the waterfall). Parietal alpha is weaker than occipital, so set Alpha thresh between your eyes-open and eyes-closed values.
- **Environment.** Upright and relaxed, screen at eye level ~60 cm; unplug the laptop charger if possible; away from mains cables and motors; phone away; no gum.
- **Dongle latency (Windows).** FTDI latency timer → 1 ms (Device Manager → COM port → Port Settings → Advanced).

## 5. Recording rules (both models)

- Stay still and silent during recorded periods — no talking, head movement or shifting.
- Blink only during the yellow *get ready* cue (MI) or the fixation cross (ErrP) — never during REST, imagery or an outcome.
- No jaw clenching or facial/neck tension: muscle activity swamps EEG.
- Keep a soft, steady gaze on the cue. Ignore the small countdown in the corner — it is deliberately dim and off-centre.
- Runs of about 3–5 minutes with breaks; fatigue degrades both signals.

## 6. Model 1 — Motor imagery (rest / left / right)

### 6.1 Why rest is a class

Imagining a hand movement desynchronises the mu (8–12 Hz) and beta (13–30 Hz) rhythms over the opposite hemisphere's hand area — C3 for the right hand, C4 for the left — and they rebound (ERS) afterwards [1, 2]. During rest they stay high. Training *rest* as its own class therefore teaches the ERD/ERS gate and the left/right decision in one model; P(rest) acts as a soft gate.

### 6.2 Subject technique

- **Imagine, don't move.** Imagine the *feeling* of slowly squeezing the cued hand, as if your own muscles did it — not a video of yourself. Hands stay relaxed and still. A real squeeze is motor execution (another signal plus muscle artefact), so a model trained on it fails on pure imagery.
- **Warm-up (not recorded):** 3–5 slow real squeezes per hand, then reproduce that feeling without moving.
- Use the **same strategy every trial** — consistency beats intensity.
- **Yellow** = prepare, don't start. **Green** = imagine for the whole block; stop when the cue changes. **Blue REST** = relax, fixate and keep still — rest is recorded data too.

### 6.3 Recording protocol

Each rep records REST, LEFT and RIGHT once each **in random order**, each after an unrecorded 2 s *get ready* cue; counts stay equal and no cue repeats back-to-back. Each recorded block is one **trial**; its first 0.4 s is dropped as cue-onset transient. A rep takes 18 s at 4 s per class, so 10 reps is a 3-minute run.

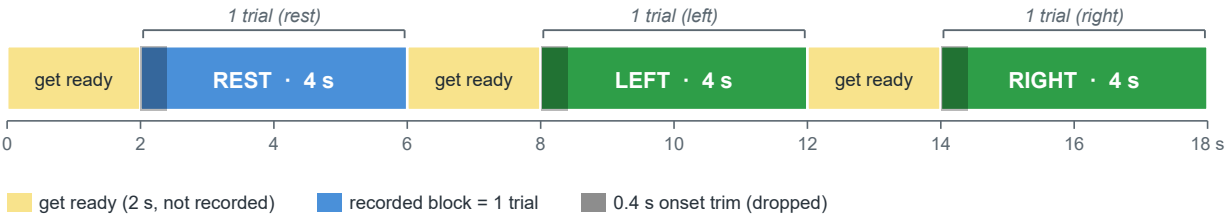



*Figure 2 — One example MI rep at the default 4 s per class (colours as on the Cue display). The order inside each rep is randomised.*

**How much:** aim for at least 25–30 trials per class — for example 3 runs of 10 reps. Cross-validation uses up to 5 folds; with fewer than 5 trials in a class it uses fewer folds, and with fewer than 2 it cannot validate at all.

### 6.4 Features and model

Raw window → band-pass 8–30 Hz → Common Spatial Patterns (CSP) → log-variance → standardise → classifier. CSP learns spatial filters that maximise the variance difference between classes [3]. With three classes it runs one-vs-rest — a CSP per class against the other two, 4 filters each — giving 12 features [4].

**Table 3 — Classifier choices (Train → Model)**

| Classifier | Type | When to use |
|---|---|---|
| **lda** | Linear discriminant (default) | Start here: robust with little data, fast; the standard MI choice [5]. |
| **svm** | Linear support-vector machine | Often matches LDA; may edge ahead with more trials. |
| **logreg** | Logistic regression (linear) | Similar boundary to LDA; well-behaved probabilities. |
| **rf** | Random forest, 200 trees | Non-linear; needs more data to beat the linear models. |
| **tree** | Single decision tree | Interpretable but tends to overfit CSP features — use for comparison only. |

**What to expect:** chance is 33%. With dry electrodes an early target is roughly 50–70% balanced accuracy, with rest vs imagery usually separating more easily than left vs right. A sizeable minority of users need several sessions before their imagery patterns separate [6] — treat these figures as indicative, not a guarantee.

### 6.5 Settings

**Table 4 — MI settings in the GUI**

| Tab | Setting | Recommended | Options / range | Notes |
|---|---|---|---|---|
| **Record** | Subject | Your ID, e.g. *amogh* | Free text | Names the folder eeg_<subject>_<date-time>. |
| | Paradigm | MI (left/right) | MI · ErrP | REST, LEFT and RIGHT blocks (3-class). |
| | Seconds/class | 4 | 2–30 s | Length of each recorded block; must exceed the Train window + 0.4 s. |
| | Reps | 10 per run | 1–15 | Each class once per rep, random order. |
| | Description | Run notes | Optional | Saved in the recording's JSON. |
| **Train** | Modality · Paradigm | EEG · MI (rest/L/R) | — | Lists only MI recordings. |
| | Model | lda | lda · svm · logreg · rf · tree | See Table 3. |
| | Win / Step ms | 1000 / 250 | ms | Training window and hop; also the model's Live window. Longer is steadier but slower to respond. |
| | Recordings | 2–3 runs, pooled | Ctrl/Shift-click; Refresh | Same montage, rate and labels required. |
| | Description | Model notes | Optional | Saved in the model's JSON. |
| **Live** | Models | The saved MI model | Multi-select; refresh; Start | Window from the model; update rate from Config Step. |

### 6.6 In the app

1. **Config** → Load MI/ErrP preset → Source *cyton* + COM port → Apply & Save.
2. **Monitor** → run the checks in section 4.
3. **Record** → Subject → Paradigm *MI (left/right)* → Seconds/class 4, Reps 10 → Description → Record session, and follow the cue. Repeat for 3 runs with breaks.
4. **Train** → EEG → *MI (rest/L/R)* → Model *lda* → Win/Step 1000/250 → Ctrl-click the runs → Train → read the report → Save model.
5. Optionally retrain the same runs with svm, logreg, rf or tree and keep the model with the best macro-F1.

## 7. Model 2 — Error-related potentials (correct / error)

### 7.1 Subject technique

- **Care about the outcome.** Watch the target, then the result, and let the natural "that's wrong" reaction happen.
- Never move, sigh or clench on an error — a movement time-locked to errors is the fastest way to ruin the data, because the model learns the artefact.
- Don't try to predict outcomes; react honestly. Blink during the fixation cross.

### 7.2 Recording protocol

A cued feedback task on the Cue display: fixation, a blue target arrow, then an outcome arrow. At the set error rate the system deliberately shows the other direction. The outcome onset is time-stamped as *correct* or *error*, and each event yields one epoch — one trial for cross-validation.

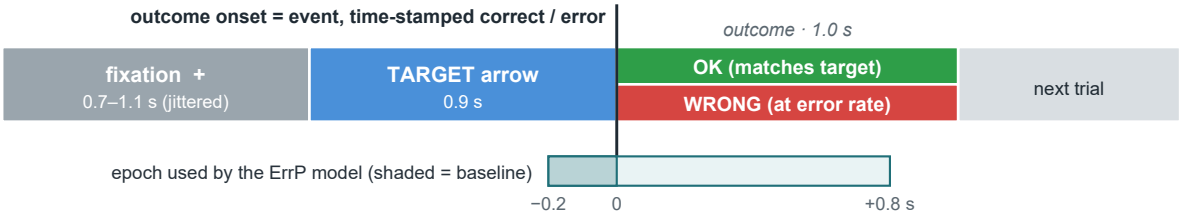



*Figure 3 — One ErrP trial and the epoch extracted around the outcome onset.*

**How much:** about 3 s per trial, so 100 trials take roughly 5 minutes. Aim for at least 200 trials (about 60 errors at 0.30) over 2 runs; 60 trials give only about 18 errors, too few for a stable 5-fold estimate.

### 7.3 Features and model

Band-pass 1–10 Hz keeps the slow error components — the error-related negativity at fronto-central sites, then the error positivity (Pe) [7, 8]. Each epoch is baseline-corrected on −0.2 to 0 s, split into 12 time bins and averaged per bin and channel (12 × 8 = 96 features), then classified by shrinkage LDA with balanced class priors — the standard single-trial ERP classifier [9].

**What to expect:** about 65–80% balanced accuracy in good setups. Each outcome is time-stamped after it is painted and mapped to the EEG through the stream clock, so buffering delays don't shift it; a frame or two of display lag remains. Be sceptical of high scores from few trials.

### 7.4 Settings

**Table 5 — ErrP settings in the GUI**

| Tab | Setting | Recommended | Options / range | Notes |
|---|---|---|---|---|
| **Record** | Subject | Your ID | Free text | As for MI. |
| | Paradigm | ErrP (feedback task) | MI · ErrP | Runs the target/outcome task on the Cue display. |
| | Trials | 100 per run | 20–400 | About 3 s per trial. |
| | Error rate | 0.30 | 0.10–0.50 (step 0.05) | Share of wrong outcomes; rarer errors evoke a stronger response, but fewer epochs. |
| | Description | Run notes | Optional | Saved in the recording's JSON. |
| **Train** | Modality · Paradigm | EEG · ErrP | — | Lists only ErrP recordings. |
| | Model | — (hidden) | Fixed | Always shrinkage LDA. |
| | Win / Step ms | — (ignored) | — | ErrP uses fixed −0.2 to +0.8 s epochs. |
| | Recordings | 2 or more runs, pooled | Ctrl/Shift-click; Refresh | Same pooling rule as MI. |
| **ErrP Live** | Test · Trials · Error rate · Threshold | Arrow · 40 · 0.30 · 0.50 | Arrow / Facts; threshold 0.05–0.95 | Each outcome scored; threshold re-scores all. Facts: section 9. |

### 7.5 In the app

1. Config preset and Monitor checks as for MI.
2. **Record** → Subject → Paradigm *ErrP (feedback task)* → Trials 100, Error rate 0.30 → Record session, and watch the screen. Repeat for a second run.
3. **Train** → EEG → *ErrP* → Ctrl-click the runs → Train → Save model.

## 8. Evaluation: how to read the report

Training never splits windows at random. Neighbouring windows from the same block are nearly identical, so a random split leaks test data into training and inflates the score [10]. Instead:

- **Grouping:** every MI block and every ErrP event is one trial, and all of a trial's windows or epochs stay in the same fold.
- **Folds:** StratifiedGroupKFold with k = min(5, fewest trials in any class); classes are balanced across folds.
- **Report:** out-of-fold predictions are pooled into the per-class report shown in the Train tab's right-hand box; the saved model is then refit on all the data.

**Table 6 — Metrics in the report**

| Metric | What it means | What to look for |
|---|---|---|
| **precision** | Correct share of a class's predictions. | Low for left/right → rest misread as a command. |
| **recall** | Share of a class's windows caught. | MI rest: gate reliability. ErrP error: errors caught. |
| **F1** | Harmonic mean of precision and recall. | Compare classifiers by macro-F1 (mean over classes). |
| **support** | Windows (MI) or epochs (ErrP) per class. | ~11 per 4 s block; if uneven, record more of that class. |
| **balanced accuracy ± std** | Mean per-class recall; mean ± std over folds. | Chance 33% (MI) / 50% (ErrP); large std → record more. |

Within-session cross-validation is still optimistic. The honest test is a new session — re-seat the headset on another day and run the saved model in Live.

## 9. Live use

- EEG → Live → refresh → select the MI model → Start. Bars show P(rest), P(left) and P(right), with the top class highlighted and repeated on the DECISION line.
- *Rest* = gate closed, no command. Live currently reports the top class at every update; a confidence threshold and smoothing for drone commands will be tuned during integration.
- The current Config montage must contain the model's electrodes (the preset guarantees this). The window comes from the model and the update rate from Config Step (minimum 150 ms).
- A **data delay** readout (Monitor, Live, ErrP Live) shows how stale the newest sample is and warns when data arrives in bursts; training and live use the same causal filters, so live results match training exactly.
- ErrP models are event-locked, so they are tested on the **ErrP Live** tab instead: it runs the feedback task, classifies each outcome ~0.8 s after it is shown and scores it against what was shown (accuracy, errors caught, false alarms, confusion matrix).
- ErrP Live's **True / false facts** test shows easy statements (100 true, 100 false) for *Fact time* (default 15 s, 2–60 s); every 0.1 s epoch is classified while one is shown and its *Score* (peak or mean P(error)) is thresholded. Untick *Show results* to hide the verdicts from the subject.

## 10. Session cadence and pooling

- Short runs with breaks; two sessions on different days beat one long session.
- Retrain each session: EEG shifts between sessions and headset placements.
- Pool runs only when montage, sample rate and labels match — the Train tab blocks mismatches. Trial IDs stay unique across pooled runs.
- Record MI and ErrP as separate sessions.

## 11. Pre-flight checklist

- ☐ A1 → SRB2 and A2 → BIAS clipped on and wired.
- ☐ Combs on the scalp; contacts low and even.
- ☐ Config: MI/ErrP preset, Source cyton + COM port, Apply & Save.
- ☐ Monitor: no RAILED/FLAT flags, blink visible at Fz, no wall of δ, alpha rises at P3/P4 with eyes closed.
- ☐ Charger unplugged or away from mains; phone away; seated upright.
- ☐ Record: Subject entered, paradigm chosen, description added.
- ☐ MI: imagine — don't squeeze; same strategy; still during REST; blink only on yellow.
- ☐ ErrP: care about the outcome; don't move on errors.

## Changes from revision 1

- ERD/ERS is no longer a separate model — it is the MI model's *rest* class.
- MI is 3-class with a selectable classifier (lda, svm, logreg, rf, tree).
- MI cues are block-randomised: equal rest/left/right counts, never the same cue twice in a row.
- *Rest* blocks are recorded and trained, so they must be as clean as imagery blocks (no blinking or fidgeting).
- Evaluation is trial-grouped, class-stratified k-fold cross-validation with per-class precision, recall, F1 and support.
- Recordings carry a *trial* column; the subject is entered in the Record tab.
- New ErrP Live tab: test a trained ErrP model on live EEG with per-trial scoring.
- Causal filtering in training and live: live results match training exactly, and the ErrP verdict arrives ~0.8 s after the outcome.
- Outcomes are mapped to the EEG through a stream clock, immune to USB buffering bursts.
- Reference (SRB) and bias must be physically connected; the alpha check runs on P3/P4.

# Appendix A — Settings fixed in code (not in the GUI)

**Table 7 — Fixed processing and timing**

| Item | Value | Where |
|---|---|---|
| **MI band-pass** | 8–30 Hz, causal (same filter live and in training) | eeg_training.MI_BAND |
| **CSP** | 4 filters per CSP; one-vs-rest for 3 classes (12 features) | eeg_models.MIModel |
| **Onset trim** | 0.4 s at the start of each recorded MI block | transient_ms in config/eeg.yaml (no GUI field) |
| **MI get-ready cue** | 2 s, not recorded | Record tab (MI) |
| **MI cue order** | Block-randomised: each class once per rep; no immediate repeats | eeg_tab.mi_cue_order |
| **ErrP band-pass** | 1–10 Hz, causal (same filter live and in training) | eeg_training.ERRP_BAND |
| **ErrP epoch / baseline** | −0.2 to +0.8 s / −0.2 to 0 s | ERRP_TMIN, ERRP_TMAX |
| **ErrP features** | 12 time bins × 8 channels = 96 | eeg_models.ErrPModel |
| **ErrP classifier** | Shrinkage LDA, balanced priors | eeg_models.ErrPModel |
| **ErrP task timing** | Fixation 0.7–1.1 s · target 0.9 s · outcome 1.0 s | Record tab (ErrP) |
| **Cross-validation** | StratifiedGroupKFold, k = min(5, fewest trials in any class); final model refit on all data | eeg_training._grouped_cv |
| **Live MI** | Continuous causal 8–30 Hz filter → newest window → model; update every ≥ 150 ms | eeg_tab.EEGEngine |
| **Event timing** | Outcome painted, then time-stamped and mapped to its sample via the stream clock | eeg_tab.EEGEngine.index_at |
| **ErrP Live verdict** | As soon as the epoch ends (~0.8 s after the outcome) | eeg_tab.EEGEngine |

# References


[1] Pfurtscheller G, Lopes da Silva FH. Event-related EEG/MEG synchronization and desynchronization: basic principles. *Clin Neurophysiol*. 1999;110(11):1842–1857.

[2] Pfurtscheller G, Neuper C. Motor imagery and direct brain–computer communication. *Proc IEEE*. 2001;89(7):1123–1134.

[3] Ramoser H, Müller-Gerking J, Pfurtscheller G. Optimal spatial filtering of single trial EEG during imagined hand movement. *IEEE Trans Rehabil Eng*. 2000;8(4):441–446.

[4] Dornhege G, Blankertz B, Curio G, Müller K-R. Boosting bit rates in noninvasive EEG single-trial classifications by feature combination and multiclass paradigms. *IEEE Trans Biomed Eng*. 2004;51(6):993–1002.

[5] Lotte F, Bougrain L, Cichocki A, et al. A review of classification algorithms for EEG-based brain–computer interfaces: a 10 year update. *J Neural Eng*. 2018;15(3):031005.

[6] Vidaurre C, Blankertz B. Towards a cure for BCI illiteracy. *Brain Topogr*. 2010;23(2):194–198.

[7] Chavarriaga R, Millán JdR. Learning from EEG error-related potentials in noninvasive brain–computer interfaces. *IEEE Trans Neural Syst Rehabil Eng*. 2010;18(4):381–388.

[8] Chavarriaga R, Sobolewski A, Millán JdR. Errare machinale est: the use of error-related potentials in brain–machine interfaces. *Front Neurosci*. 2014;8:208.

[9] Blankertz B, Lemm S, Treder M, Haufe S, Müller K-R. Single-trial analysis and classification of ERP components — a tutorial. *NeuroImage*. 2011;56(2):814–825.

[10] Varoquaux G, Raamana PR, Engemann DA, et al. Assessing and tuning brain decoders: cross-validation, caveats, and guidelines. *NeuroImage*. 2017;145:166–179.